Attendance Marker (am-oss)

UI & Feature Design Specification — Student & Faculty Mobile Applications

Document Type: UI/UX Architecture & Placement Guide	**Version:** 2.4.0 (Production Blueprint)
Target Audience: UI/UX Designers & Mobile Developers	**Platforms:** Android (Flutter), Web, Desktop
Roles Covered: Faculty & Student Roles	**Status:** Approved & Ready for Mockup Creation

1. Executive Overview & Global Design System

The **Attendance Marker** mobile client operates on a unified multi-role Flutter codebase supporting two core end-user roles: **Faculty** and **Student**. This specification details every implemented screen, exact component hierarchy, visual token specifications, interaction states, and precise button placements to guide the UI/UX design team in creating pixel-perfect Figma/Penpot mockups.

Token / Style Name	Hex / Spec	Role / Visual Context in App
Primary Navy	#1E3A8A / #0F172A	AppBar backgrounds, primary CTA buttons, elevated card borders
Accent Blue	#2563EB / #3B82F6	Active tabs, primary action highlights, radio selection indicators
Surface Dark (Canvas)	#0F172A	Dark theme scaffold background; high-contrast base for cards
Card Container	#1E293B (Dark) / #FFFFFF (Light)	Elevated cards, rounded rectangles (R=12px), elevation 1.0–2.0
Success Green	#16A34A / #22C55E	Attendance >=75%, Verified status, Present badge, Session Active
Warning Amber	#D97706 / #F59E0B	Attendance 65%-74.9%, Pending review, Partial enrollment
Critical Red	#DC2626 / #EF4444	Attendance <65%, Absent badge, Terminate session, Sync failure
Touch Targets	Min 48 x 48 dp (W x H)	All interactive buttons, icons, pills, and touchable elements

2. Shared Screens (Authentication & Settings)

2.1 Login Screen & Server Configuration Modal

Role Access: Both Faculty and Students log in via the unified login screen.

- **Top Header:** App logo (64x64 dp), App Title "Attendance Marker", Subtitle "Smart Academic Attendance Suite".
- **Top-Right Icon:** Server Settings Gear button (48x48 dp touch target). Opens the Server URL configuration bottom sheet modal.
- **Input Field 1 (Username / Reg No):** Outlined text field with leading icon (person), clear button, supporting uppercase auto-transform.
- **Input Field 2 (Password):** Obscured text field with trailing eye toggle (show/hide password).
- **Role Toggle / Auto-Detect:** Login automatically navigates to Faculty or Student dashboard based on JWT user claims.
- **Primary Action Button:** Full-width elevated button **[Sign In]** (H=50 dp, navy blue, white bold text). Displays CircularProgressIndicator during submission.
- **Server Config Modal:** Text field for Base URL (default http://127.0.0.1:8000), **[Test Connection]** outline button, **[Save & Close]** solid button.

3. Faculty Mobile Application Specification

The Faculty app contains a 4-tab Bottom Navigation Bar: **1. Dashboard**, **2. Attendance**, **3. Timetable**, **4. Profile**.

3.1 Tab 1: Faculty Dashboard Screen

- **Faculty Profile Card (Top):** Displays avatar with initials, Faculty Name (e.g., Dr. R. Sharma), Faculty Code (FAC04), Department, and Designation pill badge.
- **Active Class Slot Card:** Real-time banner showing current ongoing class. Displays Subject Name, Subject Code, Room No, Section, and Period time interval (e.g. 09:00 - 09:50).
- **Attendance Window Status Pill:** Dynamic status chip: *"Active Window" (Green)*, *"Window Opens in 12m" (Amber)*, or *"Closed" (Grey)*.

- **Primary Action Button:** Prominent **[Take / Resume Attendance]** button embedded directly in the active slot card. Navigates directly into the Attendance marking workflow.
- **Today's Schedule Stream:** Vertical list of chronological slot cards for today, each with time tag, room badge, and status indicator.
- **Error & Offline State:** Dedicated error banner with **[Retry]** button to recover seamlessly from network hiccups.

3.2 Tab 2: Faculty Attendance Roster & Marking Screen

- **Top Dual-Tab Switcher:** Segmented pill selector: **[Scheduled Slots]** | **[Manual Override / Roster]**.
- **Period Selector Dropdown:** Allows choosing among eligible assigned periods for the current day.
- **Attendance Summary Header:** 3 stat counters: Total Enrolled, Present Count (Green), Absent Count (Red).
- **Batch Action Bar: [Mark All Present]** (outlined green) and **[Reset All]** buttons.
- **Student Roster List:** Each item row shows student roll number, name, registration number, face-verification confidence badge (e.g., 94% Biometric Match), and dual toggle buttons: **[P]** (Green pill) / **[A]** (Red pill).
- **Bottom Sticky Action Bar:** Fixed bottom container with **[Finalize & Submit Attendance]** (elevated solid button, H=52 dp).

3.3 Tab 3: Faculty Weekly Timetable Screen

- **Horizontal Day Selector:** Monday through Saturday pill buttons with indicator dot for active day.
- **Timetable Slot Cards:** Chronological cards showing Start Time, End Time, Subject Code, Subject Title, Section (e.g., CSE-B), and Assigned Room Number.
- **Empty State Graphic:** "No classes scheduled for this day" illustration when faculty has a free day.

3.4 Tab 4: Faculty Profile & Account Screen

- **Profile Header:** Large circular avatar, Full Name, Employee ID, Email, Department.
- **Security Options: [Change Password]** list tile opening password change sheet.
- **System Info:** App build version, API endpoint URL, Last Sync timestamp.
- **Logout Button:** Destructive red-outlined button **[Sign Out]** with confirmation dialogue.

4. Student Mobile Application Specification

The Student app contains a 4-tab Bottom Navigation Bar: **1. Overview**, **2. Biometric Register**, **3. Timetable**, **4. Profile**.

4.1 Tab 1: Student Overview & Attendance Dashboard

- **Student Identity Card (Top):** Student avatar with initials, Full Name, Registration Number (e.g., 2024BCSE001), Department, Year, and Section (e.g. CSE-A).
- **Mandatory Password Change Alert:** If the user is logging in with initial default password, a high-priority amber warning banner renders at the top with a direct **[Update Password Now]** button.
- **Overall Attendance Radial Gauge Card:** Circular or semi-circular progress meter displaying aggregate attendance percentage (e.g. 84.5%). Color transitions dynamically: Green (>=75%), Amber (65%-74.9%), Red (<65%). Displays required classes needed to reach 75% threshold.
- **Quick Metric Counters (Grid 2x2):** Total Conducted Classes, Total Attended Classes, Total Absent, Approved On-Duty (OD) credits.
- **Subject-Wise Attendance Breakdown:** Card list for all enrolled subjects. Each card includes: Subject Code (e.g. CS301), Subject Name, Percentage Progress Bar, Fraction (Attended/Total, e.g. 28/32), and Minimum 75% Status Badge.

4.2 Tab 2: Biometric Multi-Pose Face Enrollment Screen

- **Guidance Instruction Banner:** Header detailing camera alignment requirements (clean lighting, neutral expression, no sunglasses/masks).
- **Pose Progress Stepper (3-Stage):** 1. Frontal Pose (0° ±5°) → 2. Left Angle (15°) → 3. Right Angle (15°).
- **Live Camera Viewfinder / Preview:** Centered elliptical face guide overlay with real-time feedback bounding box. Bounding box turns Green when face is well-centered and lighting is sufficient.
- **Quality & Angle Telemetry:** Displays real-time Yaw, Pitch, Roll angles and face clarity score.
- **Primary Action Button:** Floating / Bottom-centered **[Capture Pose Photo]** (Circular camera icon button, 64x64 dp).
- **Validation Review Modal:** Shows thumbnail of captured pose with quality metrics and dual actions: **[Retake]** (Outline) and **[Confirm & Next Pose]** (Solid Green).
- **Final Submission Button:** Once all 3 poses are captured, activates **[Submit Face Templates for Approval]**.

4.3 Tab 3: Student Class Timetable Screen

- **Day Filter Selector:** Horizontal scrollable pills for Mon, Tue, Wed, Thu, Fri, Sat.
- **Period Time-block Cards:** Displays chronological period slot: Time (08:30 - 09:20), Course Name, Course Code, Faculty In-charge name, and Classroom/Lab Room number.
- **Ongoing Class Highlighting:** The currently running period card has an elevated border and a pulsating "In Progress" badge.

4.4 Tab 4: Student Profile & Settings Screen

- **Student Profile Header:** Student photo or monogram initials, Full Name, Reg No, Roll No, Official Email, Academic Batch.
- **Face Enrollment Status Tile:** Shows whether biometric templates are *Registered & Verified (Green)* or *Pending Registration (Amber)* with a direct **[Re-enroll Face]** action.
- **Account Management: [Change Password]** action tile opening password modal with validation criteria.
- **Logout Button:** Destructive red-outlined **[Log Out]** button at the bottom of the scroll view.

5. Button Placement & Interaction Specification Matrix

The following table defines the exact screen placement, dimensions, visual styling, and behavioral transitions for every interactive button across both applications.

Screen / Context	Button Label	Exact UI Placement	Styling & Dimensions	Target Action / Outcome
Login Screen	[Sign In]	Below password field, full-width with 24dp horizontal margin	Solid Navy Blue (#1E3A8A), H=50dp, R=10dp, White bold text	Authenticates user, redirects to role dashboard
Login Screen	[Server Config]	Top-right AppBar action icon (Gear icon, 48x48dp target)	Icon-only, Slate-400 color, transparent background	Opens modal bottom sheet to configure backend host URL
Faculty Dashboard	[Take / Resume Attendance]	Bottom of Active Class Slot Card, aligned right / full-width	Solid Emerald Green (#16A34A), H=44dp, R=8dp, White text	Opens Attendance Marking screen for the current slot
Faculty Dashboard	[Retry]	Inside network error card, centered below error message	Outlined button with blue border, H=38dp, R=6dp	Triggers full reload of dashboard schedule & session data
Faculty Attendance	[Mark All Present]	Above roster list, right side of batch action bar	Outlined Emerald Green, H=36dp, R=6dp, Green text	Sets all student attendance toggles to Present in memory
Faculty Attendance	[P] / [A] Toggles	Right side of each student roster list item	Pill toggle pair (36x32dp each), Green for P, Red for A	Toggles individual student attendance status
Faculty Attendance	[Submit Attendance]	Sticky fixed bottom bar across screen bottom	Full-width solid Navy (#1E3A8A), H=52dp, R=0 (or 12dp sheet)	POSTs finalized attendance payload to backend API
Student Dashboard	[Update Password]	Inside mandatory password alert card, right aligned	Solid Amber (#D97706), H=36dp, R=6dp, White text	Opens password change bottom sheet modal
Student Biometric	[Capture Pose Photo]	Bottom-center of camera viewfinder, 24dp above screen edge	Circular shutter button, 68x68dp, White ring with blue core	Captures photo, triggers on-device pose & quality analysis
Student Biometric	[Confirm & Next]	Inside pose review bottom sheet, right side	Solid Green (#16A34A), H=46dp, R=8dp, White text	Stores current pose template and advances to next pose
Student Biometric	[Retake Pose]	Inside pose review bottom sheet, left side	Outlined Grey/Red, H=46dp, R=8dp	Discards current capture and reactivates camera stream
Student Profile	[Re-enroll Face]	Inside Face Registration tile, right aligned	Outlined Blue button, H=34dp, R=6dp	Navigates directly to multi-pose biometric enrollment flow
Profile (Both)	[Sign Out]	Bottom of Profile screen scroll view, centered / full-width	Outlined Crimson Red (#DC2626), H=48dp, R=8dp	Clears JWT tokens, user cache, and navigates to Login

6. UI Designer Implementation Guidelines & Deliverables

To ensure seamless development handoff, the UI designer should adhere to the following conventions when producing Figma / Penpot design assets:

- **Grid & Layout:** 8dp grid system. Standard screen padding = 16dp horizontal, 12dp vertical. Card internal padding = 16dp.
- **Component Modularization:** Create reusable component variants for: Primary Buttons, Status Badges (Green/Amber/Red/Grey), Stat Counters, Schedule Cards, and Roster Item Rows.
- **Dynamic States:** Provide mockups for *Default*, *Loading (shimmer skeleton)*, *Empty state*, and *Error with retry* for all dashboard and timetable screens.
- **Dark / Light Theming:** The Flutter codebase implements dark mode by default (`#0F172A` background, `#1E293B` cards). Designer should provide both Dark Mode and Light Mode (`#F8FAFC` background, `#FFFFFF` cards) color styles.
- **Accessibility (a11y):** Minimum contrast ratio of 4.5:1 for all body text against card backgrounds. Minimum touch target size of 48x48dp for all interactive icons and buttons.